UNIVERSIDAD INDUSTRIAL DE SANTANDER
Escuela de Ingeniería Eléctrica, Electrónica y de Telecomunicaciones

Advanced Layout and Sensitivity Analysis of a Voltage Comparator in 28 nm CMOS Technology

Presented To:
DEGREE WORKS COMMITTEE E^{3}T

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Universidad Industrial de Santander
Escuela de Ingeniería Eléctrica, Electrónica y de Telecomunicaciones
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Professors
DEGREE WORKS COMMITTEE
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Unversidad Industrial de Santander
Present

Reference: Work plan in the modality of research project: "Advanced Layout and Sensitivity Analysis of a Voltage Comparator in 28 nm CMOS Technology".

Dear professors,

Considering the articles 3o., 8o. and 11th. of chapter IX of title V of the undergraduate student academic regulations, we present for your consideration the degree plan in the form of a research project: Advanced Layout and Sensitivity Analysis of a Voltage Comparator in 28 nm CMOS Technology." prepared by the electronic engineering students Michel Yurany Saenz Lopez and Duvan Nicolas Gomez Diaz, codes 2205602 and 2202786. This document has our approval, for which we respectfully request your evaluation.

Best regards,

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SUMMARY OF THE PROPOSED THESIS

Title:	Advanced Layout and Sensitivity Analysis of a Voltage Comparator in 28 nm CMOS Technology.
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GENERAL OBJECTIVE

To enhance the robustness and performance of a Voltage Comparator by applying advanced layout techniques and sensitivity analysis in 28 nm CMOS technology.

PROJECT DESCRIPTION

The project aims to improve the robustness and performance of a voltage comparator through the application of advanced layout techniques and sensitivity analysis in 28 nm CMOS technology. Starting from a baseline design, its behavior is evaluated using PVT and Monte Carlo simulations. Based on this reference, the design will be enhanced with a new layout that applies strategies to mitigate layout-dependent effects (LDE). Finally, a post-layout verification and a robust comparison with established metrics, observing the advantages of using advanced layout techniques to mitigate LDE. This contributes to strengthening the technical capabilities of the research group.

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1 Introduction

In the current era, the vast majority of electronic devices heavily rely on the energy provided by batteries. Particularly noteworthy are System-on-Chip (SoC) devices, which play a fundamental role in the functionality and efficiency of these devices. Power consumption has become a critical factor, directly impacting the battery life and active time of electronic devices. As a result, power management has gained increasing importance in the design and operation of these devices.

SoC is composed of various circuit blocks that require different voltage levels to operate at their maximum performance. These voltage levels are derived from the main power source, which in most cases is a battery. Therefore, it is necessary to have a system that can efficiently and with high-quality power supply the different voltage levels required to power the various components of the system.

In this context, DC-DC converters play an essential role. These devices are capable of efficiently modifying the voltage level, either reducing or increasing it, from a fixed level provided by the power source. However, to achieve high levels of efficiency and power quality, these converters need to be controlled by a system capable of measuring the output and providing feedback to the input. This feedback allows for correcting any variations in the output and ensuring a constant and appropriate voltage level.

This project focuses on improving the design of a voltage comparator using advanced layout tech niques and sensitivity analysis to mitigate layout-dependent effects (LDE) for a PWM (pulse width modulation) integrated into a DC-DC step-down converter. The comparator plays an important role as it generates a PWM signal, which is crucial in regulating the output voltage, efficiency, and power quality of electronic devices that depend on this converter.

2 Problem Definition

The central problem addressed falls within the scope of energy management in microelectronics, specifically in the context of analog design and the efficient operation of electronic devices and integrated systems. Nowadays, it is essential that these devices minimize their energy consumption when they are not in use or when lower operating power is required.

The DC-DC converter is one of the most important elements in power conversion. The efficiency and accuracy of the output voltage regulation of this converter largely depend on the associated PWM controller. Therefore, the main objective of this research is to improve the design of the voltage comparator that provides the PWM control signal [2].

During the problem definition phase, it is essential to identify and consider the criteria that may affect the behavior of the voltage comparator, since the comparator must be able to respond quickly and accurately to variations, ensuring the generation of a stable PWM signal. The most important variables to be considered in this research project are the input voltage, load current, switching frequency, temperature conditions, and process and voltage variations. Likewise, since layout-dependent effects (LDE) can introduce variations in the devices and affect the comparator's accuracy, the implementation of the layout is considered.

2.1 Problem Variables

During the problem definition phase, it is essential to identify and consider the variables and criteria that can affect the behavior of the voltage comparator, as it must be able to respond quickly and accurately to variations, ensuring the generation of a stable PWM signal. The main variables considered in this research are:

- Input Voltage: The input voltage must comply with a common mode input range. Variations in this parameter can cause instability in the comparator's reference levels and affect the accuracy of the PWM signal.

- Load Current: Refers to the current drawn by the load connected to the converter. Changes in load current can alter the comparator's operating point and affect regulation performance.

- Switching Frequency: Determine the switching rate of the PWM controller operates. A higher frequency switching can improve transient response but also increases power losses and noise coupling in the comparator.

- Temperature Conditions: Temperature variations influence the electrical characteristics of semiconductor devices, such as mobility and threshold voltage, which can lead to offset errors and propagation delays in the comparator.

- Process and Voltage Variations (PVT): These are inherent manufacturing and operating deviations that can affect the stability and delay of the comparator. Robustness against PVT variations is crucial to guarantee performance consistency.

2.2 Design Criteria

To ensure optimal performance of the voltage comparator and overall efficiency of the DC-DC converter, the following design criteria are established:

- Supply Voltage: Typically 1.8 V, with a variation of ±10%, compatible with modern integrated systems.

- Power Consumption: Minimizing static and dynamic power consumption is critical in microelectronic systems. The comparator must operate efficiently without compromising performance, with a maximum power consumption of 60 uW.

- Current Bias: The bias current for the comparator is fixed at 1 uA. The bias current defines the operating point of the comparator's transistors.

2.3 Consideration of Standards and Norms

Although the project is primarily academic, it adheres to the professional and ethical standards required in analog environments. All simulations, layouts, and documentation will comply with the specifications and verification procedures of the TSMC 28 nm CMOS Process Design Kit (PDK), including Design Rule Check (DRC) and Layout Versus Schematic (LVS) validations. Additionally, all technical documentation and references will follow IEEE formatting and citation standards to ensure consistency and traceability. These considerations guarantee that the design process remains aligned with engineering norms and academic integrity, even though no physical fabrication will be performed.

2.4 Consideration of Non-Technical Aspects

Although this project is primarily technical and focused on the design of analog integrated circuits, a brief assessment was conducted to identify any potential environmental, social, health, safety, or economic implications derived from its execution. The design and simulation activities will be carried out entirely using digital tools—such as Cadence, Synopsys, and related EDA software—within the facilities of the Universidad Industrial de Santander (UIS), without the use of physical materials, chemicals, or manufacturing processes. Therefore, no direct environmental or health risks are expected. From an economic standpoint, the project contributes to the efficient use of institutional resources by leveraging existing software licenses and computational infrastructure. Socially, it promotes technological capacity-building in advanced CMOS design and strengthens local academic research aligned with global semiconductor trends. In summary, the study concludes that, while the project does not generate adverse non-technical impacts, it contributes positively to education, innovation, and technological development within the university context.

3 Restrictions

It is important to note that the design of a voltage comparator in 28 nm CMOS technology presents challenges and constraints such as:

- Minimum power supply: The voltage comparator must be designed with a fixed power supply, as it is derived from the DC-DC converter and is not modifiable.
- The available devices of the PDK (Process Design Kit) in the CMOS technology to be used: The voltage comparator will be designed using 28 nm CMOS technology, taking into account its limitations and characteristics.
- Post-layout simulations: After layout implementation, parasitic capacitances and resistances significantly alter circuit performance. Therefore, post-layout simulations are mandatory to verify and optimize the comparator's real behavior before fabrication.

Therefore, it is essential to consider the limitations of this technology throughout the entire circuit design process. This involves adhering to specific design rules (DRC and LVS) and verifying functionality, while also accounting for the various parasitic effects that arise during post-layout simulation.

4 General Objective

- To enhance the robustness and performance of a Voltage Comparator by applying advanced layout techniques and sensitivity analysis in 28 nm CMOS technology.

5 Specific Objectives

- Review and characterize the baseline Voltage Comparator design measuring robustness through PVT and Monte Carlo simulations .
- Study and document the main advanced layout techniques to mitigate layout dependent effects (LDE) in advanced nodes.
- Redesign the layout using advanced layout techniques.
- Perform post-layout verification and comparison with the baseline design including PVT and Monte Carlo simulations.

6 Preliminary studies for Plan formulation

In the preliminary studies, a bibliographical exploration oriented to the operation, the study of the voltage comparator topologies and layout-dependent effects (LDEs) is carried out, also a study of the design rules of the 28nm CMOS technology is carried out in order to solve the stability and sensitivity problems of the voltage comparator ensuring a robust design. These activities were complemented by discussions with professors and researchers in the field of microelectronics, who emphasized the need to strengthen institutional capabilities in analog IC design using advanced CMOS nodes. This understanding guided the project toward a solution that not only fulfills academic objectives but also contributes to the university's mission of fostering innovation, technical expertise, and sustainable technological development in the national semiconductor context.

6.1 Analysis of Problem Causes

The main cause addressed by this research is the instability generated by layout-dependent effects (LDEs) in 28nm CMOS technologies. As transistors become smaller, they are affected by complex physical phenomena that directly impact circuit design, limiting its stability and robustness. The project directly addresses this problem by adopting mitigation techniques for layout-dependent effects and designing them using sensitivity analysis in the voltage comparator.

6.2 Ideation Process

During the ideation phase, several alternative approaches were explored to address the challenges related to stability, sensitivity, and energy efficiency in the design of the voltage comparator. The divergent ideation stage encouraged the formulation of multiple design concepts, ranging from conventional static comparators with offset compensation techniques to more audacious proposals such as dynamic comparators with adaptive biasing and hybrid architectures that mitigate layout-dependent effects (LDEs) through symmetry and common-centroid strategies. After evaluating the technical feasibility, complexity, and potential performance of each alternative, as well as considering non-technical aspects such as resource availability, environmental sustainability, and alignment with the university's academic interests, a convergent ideation process led to the selection of the most balanced solution. The chosen approach combines a dynamic comparator topology optimized for low power consumption with a layout design strategy that minimizes mismatch and parasitic effects, ensuring robust performance in 28 nm CMOS technology. This solution was considered the most convenient and sustainable in terms of energy efficiency, institutional capabilities, and contribution to technological innovation.

6.3 Layout depent effects (LDE)

Layout-Dependent Effects (LDE) are variations in the electrical behavior of devices caused by their physical layout and proximity in the integrated circuit. Phenomena such as WPE, EM, STI stress, among others, alter key parameters of electronic devices, affecting the accuracy and symmetry of the circuit. These phenomena and ways to mitigate them will be discussed below.

- Antenna Effect: The antenna rules refer to the accumulation of charge in long interconnects during the reactive ion etching process, when the upper layers are not yet electrically connected to the silicon or ground. This charge, generated during intermediate fabrication stages, can build up on floating nodes and induce leakage currents that permanently damage the transistor gate oxide, compromising the reliability of the integrated circuit.

  To mitigate antenna effects, two main techniques are commonly used: jumpers and leakers.

Jumpers consist of using metal bridges that interrupt the continuity of polysilicon conductors near the gate terminals, thereby preventing antenna formation during the etching process, since the metal layers are not yet deposited at that stage. Leakers, on the other hand, rely on attaching a diffusion region to the conductor, which acts as a diode that safely drains the accumulated charge, preventing gate oxide damage and improving the reliability of the integrated circuit. [3]

- Latchup: Latch-up is an undesirable phenomenon that creates a low-resistance path within the circuit, caused by the interaction between two complementary parasitic bipolar junction transistors, QNPN and QPNP, formed by an n-p-n-p doping sequence. This effect typically occurs in configurations where NMOS and PMOS transistors are placed adjacent to each other, with their backgate and source terminals connected to ground and the power supply, respectively. When latch-up is triggered, an excessive current may flow, which in the worst case can destroy the integrated circuit.

  To mitigate the latch-up phenomenon, two main techniques are commonly used: increasing spacing and implementing guard rings. Enlarging the distance between NMOS and PMOS transistors increases the parasitic base width of the QNPN transistor, thereby reducing the likelihood of a low-resistance path being triggered. Guard rings, in turn, serve two main purposes: they provide a low-resistance path to collect injected minority carriers before they reach the base of the parasitic BJTs, and they reduce the effective substrate (Rsub) and well (Rwell) resistances, making it more difficult for injected currents to generate the voltage (0.7 V) required to forward-bias the parasitic base-emitter junctions. [3]

- Shallow Trench Isolation (STI): This effect is not caused by fabrication tolerances but rather by the mechanical stress that trench isolation regions exert on the adjacent silicon. The magnitude of this stress is temperature-dependent due to the different coefficients of thermal expansion (CTEs) of silicon and oxide. This pressure increases hole mobility while reducing electron mobility, leading to mobility variations that can cause significant mismatches in circuit performance. [4]

- Well Proximity Effect (WPE) is a phenomenon that can significantly impact transistor performance and, in some cases, even affect manufacturing yield. During the ion implantation process, dopants can scatter within the photoresist and spread randomly, leading to a higher concentration of implanted ions near the well boundaries. This non-uniform distribution results in an uneven doping profile, where transistors located close to the well edge exhibit a higher dopant concentration than those placed farther away. Consequently, the threshold voltage (V_{TH}) of MOSFETs is altered, introducing device mismatches and degrading the expected electrical performance of the circuit.
  To mitigate these effects we use the following techniques: increased spacing which involves increasing the distance between critical analog transistors and the nearest well edge, and placing non-functional dummy transistors or dummy active area structures along the well edge [4].

- Electro Migration (EM) refers to the gradual displacement of metal atoms within a conductor caused by momentum transfer from high-density electron flow. Under prolonged current stress, these electrons impart sufficient kinetic energy to the metal lattice atoms, forcing them to migrate in the direction of the electron movement. This atomic transport leads to the formation of voids in regions where atoms are depleted and hillocks where material accumulates, ultimately resulting in open circuits or short circuits. The severity of electromigration increases with higher current densities and elevated temperatures, making it a critical reliability concern in modern interconnects, especially in advanced CMOS technologies with

narrow metal lines and aggressive scaling.
Electromigration cannot be prevented from occurring in metallic routing wires; it can only be offset or restricted in its effect. This can be perform by: reservoir, via configurations, via arrays and corner bends [4].

- IR drop is an effect related to EM. If a metal wire is too thin, its resistance will be high, resulting in a higher current density (prone to EM effects) and a significant IR drop [4].

6.4 Characterization of a Comparator

To illustrate the characteristics of a comparator, the amplifier shown in Figure 1 will be considered, where V_P is the positive input of the comparator. Applying a positive voltage to V_P will cause the comparator's output to become positive. V_N is the negative input of the comparator, and applying a positive voltage to V_N will cause the comparator's output to become negative. The upper and lower voltage limits of the comparator's output are defined as V_{OH} and V_{OL}, respectively [5].

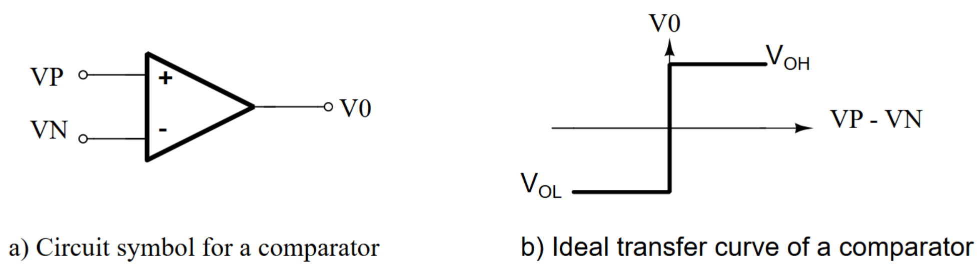

Figure 1: Circuit and Ideal transfer curve of a comparator.

6.5 Static Characteristics

An ideal comparator was previously defined. Although this type of behavior is impossible in a real situation, it can be modeled using ideal circuit elements with mathematical descriptions, which allow us to understand the characteristics of a comparator in a simpler way. Figure 2 shows a model of this ideal circuit.

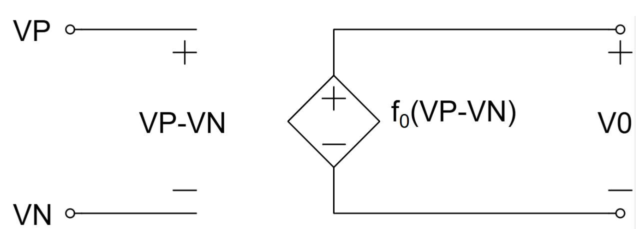

Figure 2: Model for an ideal comparator.

Analyzing Figure 1a, the comparator output is high (V_{OH}) when the difference between the non-inverting and inverting inputs is positive, and low (V_{OL}) when this difference is negative. The output changes state in response to a change in the input voltage ΔV, where ΔV tends toward zero.
This implies an infinite gain, as shown in the following equation:

$$A_v = \lim_{\Delta V \to 0} \frac{V_{OH} - V_{OL}}{\Delta V} \tag{1}$$

Once the operation of an ideal comparator has been reviewed, we analyze the behavior of a first-order model. Figure 3 shows the DC transfer curve of this model.

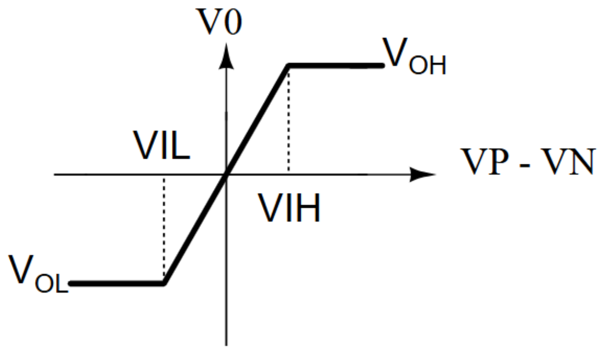

Figure 3: Transfer curve of a comparator with finite gain.

where V_{IH} and V_{IL} represent the input voltage difference required to obtain an output at its upper and lower limits, respectively. The gain of this model can be expressed as follows:

$$A_v = \frac{V_{OH} - V_{OL}}{V_{IH} - V_{IL}} \tag{2}$$

This feature is very important because it describes how the comparator works and defines the minimum amount of input change (resolution) required for the output to oscillate between the two binary states.

A non-ideal effect is the input-offset voltage V_{OS}. This phenomenon occurs when the comparator output changes state not exactly at the point where the input voltage difference is the minimum necessary for switching, but rather when this difference reaches a finite value of $+V_{OS}$. This voltage represents the shift necessary for output switching to occur. Figure 4 illustrates offset in the transfer curve for a comparator

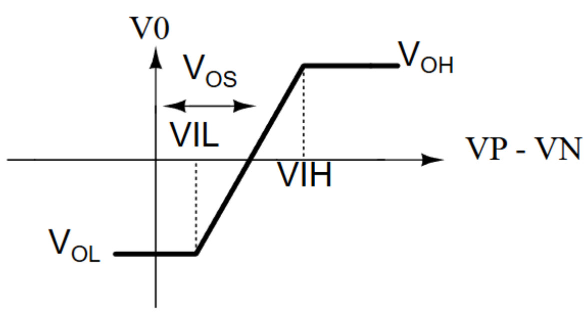

Figure 4: Transfer curve of a comparator including input-offset voltage.

This effect would not be an issue if it were predictable; however, it varies randomly from one circuit to another for a given design. Another non-ideal effect to consider is noise. This noise is modeled as if the comparator were biased within that transition region in the voltage transfer characteristic, generating uncertainty in the exact switching point as shown in Figure 5. This uncertainty can manifest itself as jitter or phase noise in systems where the comparator is used, affecting the temporal accuracy and stability of the circuit. Although a comparator is not designed to operate in the transition region, this noise becomes relevant in circuits where the comparator is used.

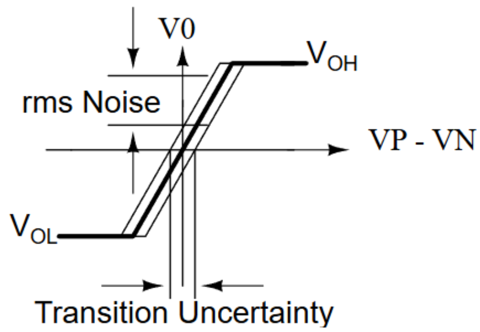

Figure 5: Influence of noise on a comparator.

6.6 Dynamic Characteristics

The dynamic characteristics of the comparator define how the comparator responds over time to a differential input signal. These characteristics include both small-signal and large-signal behavior [5]. The comparator has a propagation delay between the differential input signal and the output transition. This delay depends on the input amplitude: the greater the amplitude, the shorter the delay, until a limit defined by the slew rate (large-signal behavior) is reached.
The small-signal behavior is characterized by the frequency response of the comparator, where it can be modeled as a differential voltage gain A_V, which is expressed as:

$$A_v(s) = \frac{A_v(0)}{\frac{s}{\omega_c} + 1} = \frac{A_v(0)}{s\tau_c + 1} \tag{3}$$

The propagation delay of a comparator can be described by an exponential response when applying a differential input step. Figure 6 shows this behavior.

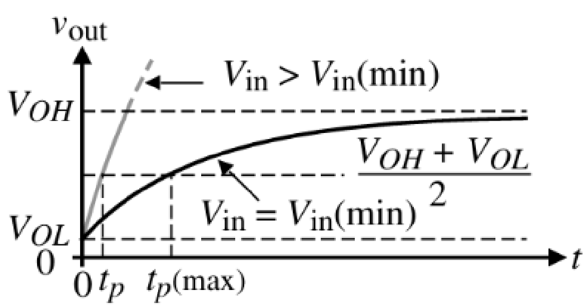

Figure 6: Small-signal transient response for a comparator.

If V_{in} is larger than $V_{in(\text{min})}$ (minimum voltage change at the comparator input), the output rise (or fall) time is faster. Therefore, it can be said that by applying a $V_{in(\text{min})}$ input we will obtain a maximum propagation delay time.
When an overdrive is applied to the input which makes V_{in} greater than $V_{in(\text{min})}$, the propagation delay time depends on the amplitude of the applied signal: the greater the input overdrive, the shorter the propagation time. However, when the overexcitation is sufficiently large, the comparator stops behaving in small-signal mode and enters large-signal mode. In this condition, the limitation is no longer imposed by the propagation delay time, but by the slew rate of the circuit. This means

that the propagation time is governed by the maximum charge or discharge speed of the internal capacitances. Therefore, improving the sink or source current capability of the comparator is key to reducing the propagation time in this regime.

6.7 Review the topology of the proposed solution

A review of the literature on the three-stage high-speed comparator topology was carried out, following the analog design methodology proposed by [1], since this was the previously chosen optimal solution topology.

This comparator consists of three stages. The first is the pre-amplification stage, which amplifies the input signal to improve the comparator's sensitivity and isolates the input from the switching noise from the comparator's positive feedback stage. The positive feedback stage determines which of the input signals is larger. The output buffer amplifies this information and outputs a digital signal [6].

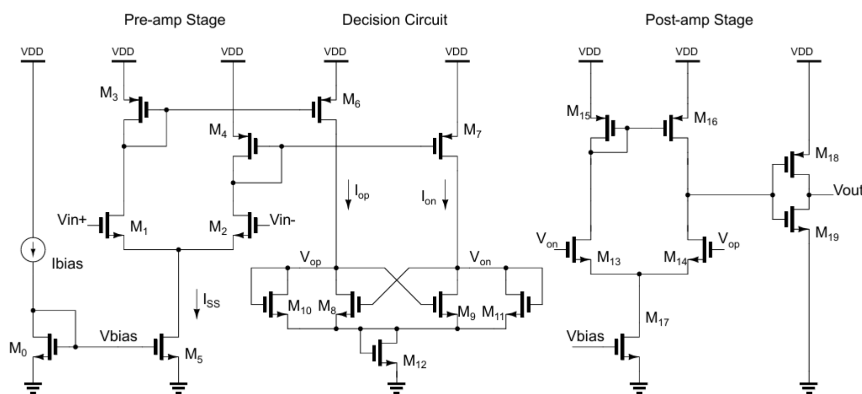

Figure 7: Comparator topology [1].

- **Pre-amp Stage**: The circuit of Figure 8 is a differential amplifier with active loads. The sizes of M_1 and M_2 are set by considering the differential amplifier transconductance, g_m, and the input capacitance. The transconductance sets the gain of the stage, while the input capacitance of the comparator is determined by the sizes of M_1 and M_2 [6].

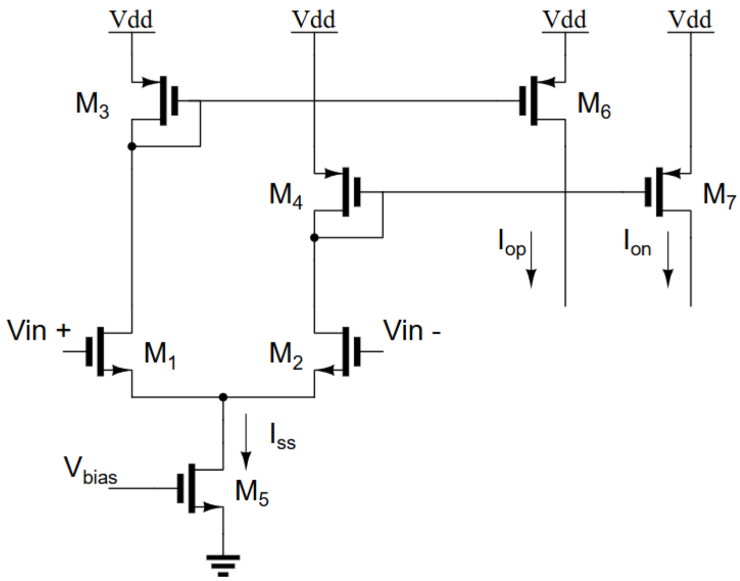

Figure 8: Preamplification stage.

Note that there are no high-impedance nodes in this circuit, other than the input and output nodes. This reduces slow internal time constants and allows for fast switching. Furthermore, using active loads (PMOS current mirror) allows for relatively high r_{out} gains without introducing large physical resistors (which would be slower and less compact).
Now we can relate the small-signal output currents i_{op} and i_{om} with the input voltages by:

$$i_{op} = \frac{g_m}{2}(v_p - v_m) + \frac{I_{SS}}{2} = I_{SS} - i_{om} \tag{4}$$

- **Decision Circuit**: The decision circuit is the heart of the comparator and should be capable of discriminating between millivolt-level signals. Figure 9 shows the circuit used in the comparator under development, which uses positive feedback through the cross-connection of gates M_6 and M_7, allowing an increase in the gain of the decision element. In addition, it is desirable to incorporate hysteresis into the design in order to suppress the effect of noise at the input.

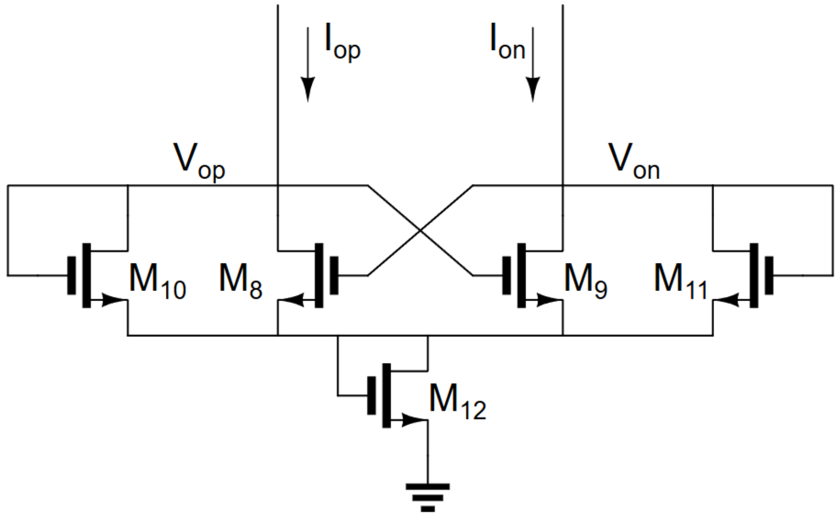

Figure 9: Positive feedback decision circuit.

Assuming that i_{op} is much larger than i_{om}, so that M_5 and M_7 are on and M_6 and M_8 are off. We also assume that $\beta_5 = \beta_8 = \beta_A$ and $\beta_6 = \beta_7 = \beta_B$, so that v_{om} is approximately 0 V and v_{op} is

$$v_{op} = \sqrt{\frac{2i_{op}}{\beta_A}} + V_{THN} \tag{5}$$

The current flowing in M_7 is

$$i_{om} = \frac{\beta_B}{2}(v_{op} - V_{THN})^2 \tag{6}$$

And the current flowing in M_5 is

$$i_{op} = \frac{\beta_A}{2}(v_{op} - V_{THN})^2 \tag{7}$$

Note that M_5 and M_7 form a current mirror, therefore we can write that

$$i_{op} = \frac{\beta_A}{\beta_B} i_{on} \tag{8}$$

If $\beta_A = \beta_B$ the switching takes place when $i = i_{om}$. The differences in β_S cause hysteresis in the comparator.

The switching voltages can be obtained by performing the following equation

$$V_{SPH} = v_p - v_m = \frac{I_{SS}}{g_m} \cdot \frac{\frac{\beta_B}{\beta_A} - 1}{\frac{\beta_B}{\beta_A} + 1} \quad \text{for} \quad \beta_B \geq \beta_A$$

$$V_{SPL} = -V_{SPH}$$

Now, as the V_{GS} of M8 increases, exceeding V_{THN}, M6 draws current from M5, which decreases the source-drain voltage of M5 and M6, causing M7 to turn off.

This topology is further explored in Behzad Razavi's paper titled "The Cross-Coupled Pair—Part I" [7]

- **Post-amp Stage**: The main purpose of the output buffer is to convert the output of the decision circuit into a logic signal. The output buffer should accept a differential input signal and not have slew-rate limitations.

  Once each stage of the comparator topology has been reviewed, the next step focuses on establishing a systematic methodology for the complete analog design process. The following section details this methodology.

7 Methodology and Schedule

7.1 Methodology

The methodology shown in Figure 10 is based on the analog design process described in [8], which was used as a reference to structure the analog design process required to meet the established objectives. First, the system defines the review and characterization of the basic design of the voltage comparator. In the next step, through PVT and Monte Carlo simulations, the robustness of the redesigned circuit is evaluated, ensuring compliance with the specifications before proceeding to the next phase. Once the design has been improved, the main advanced layout techniques to mitigate layout dependent effects (LDE) are studied and documented, and using these advanced techniques, the layout is redesigned. In the following phase, post-layout verification and comparison with the reference design are performed, including PVT and Monte Carlo simulations. Finally, the silicon fabrication and post-fabrication testing phase is carried out; however, it is important to note that this last step is beyond the scope of the current project.

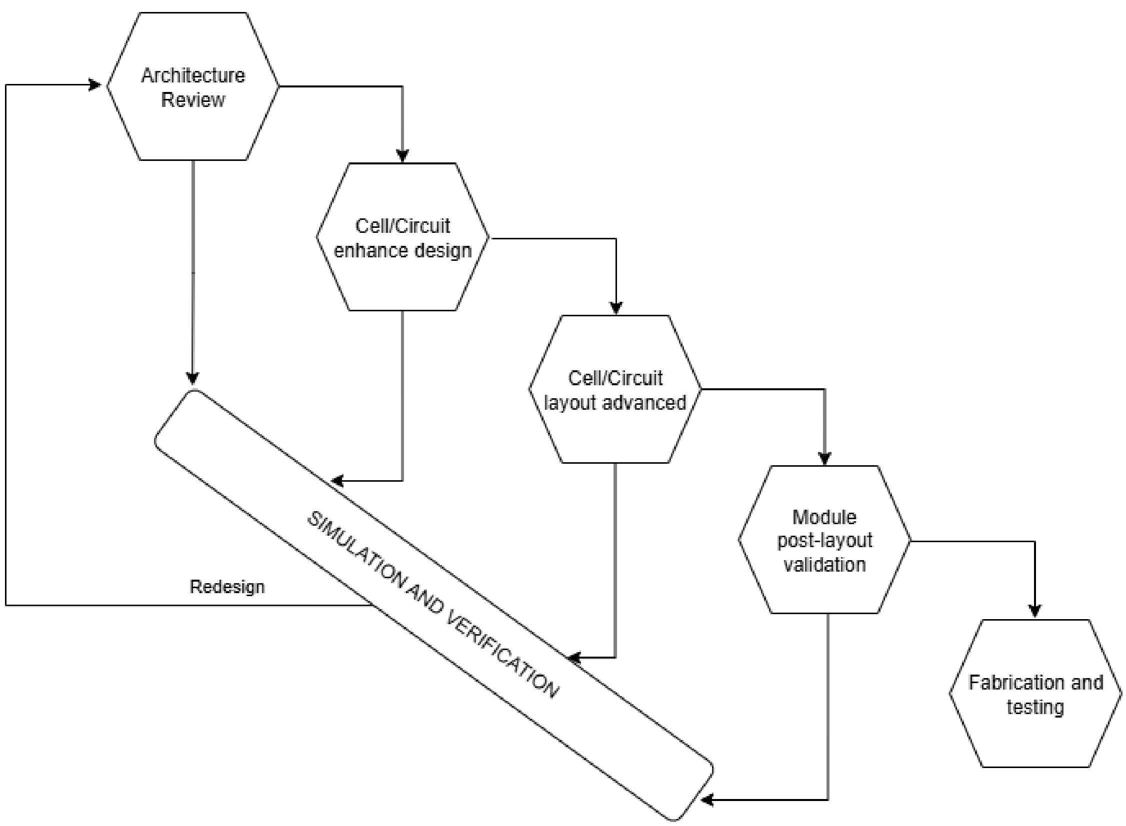

Figure 10: Visual representation of the design process.

7.2 Schedule

The plan will consist of 3 principal steps:

A. Literature review and definition of specifications

A literature review is conducted to analyze the base topology and the design aspects of each component of the voltage comparator. Furthermore, within this phase, a crucial aspect is the definition of the specifications. This process is a fundamental part of our research work, as it lays the foundation for a comprehensive understanding of the project requirements, ensuring an improvement over the baseline design.

B. Comparator voltage Design and Validation

Once the operation of the voltage comparator has been studied, the redesign and validation phase in 28 nm CMOS technology begins. The first step is to create a circuit model that can be simulated. This phase involves several activities, including:

- Design of the voltage comparator using software in the previously mentioned 28 nm CMOS technology node.
- The functionality of the voltage comparator is validated by simulating its performance using the design developed in the previous phase of the project. The simulations are carried out in the software environment to evaluate its behavior, verify its functionality, and ensure that it meets the design specifications.
- The layout development follows the design rule check (DRC) guidelines, applying advanced design techniques specific to the 28 nm CMOS process.
- Post-layout verification and comparison with the reference design are carried out, including PVT and Monte Carlo simulations to detect and measure any deviation or variation from the initially defined specifications.

C. Conclusions and document generation

The analysis of the validation results is carried out to ensure the correct operation of the system. The design of the voltage comparator must comply with all design rules to facilitate the fabrication process in future projects. Throughout the development of the project, a document will be compiled to record the results and conclusions obtained. This documentation will serve as a resource for future design projects.

Table 1 presents the Gantt Diagram of this project, which indicates the duration of each previously mentioned activity. Each column under a Month's name represents a week.

Table 1: Gantt Diagram showing the time planning of the project.

Activities	**2026**																							
	Month 1				**Month 2**				**Month 3**				**Month 4**				**Month 5**				**Month 6**			
A																								
B																								
C																								

8 Budget and Resources

This section provides information on the economic budget and necessary resources for implementing the project.

Table 2 presents an overview of the budget, including the costs associated with the Director and Co-Director, which are considered part of their academic responsibilities. The expenses incurred by the University (Universidad Industrial de Santander, UIS) and the author are assumed to be requirements for obtaining the degree (refer to Table 3). The equipment used for the project is owned by the Integrated Circuits Research Group Onchip at the Universidad Industrial de Santander. Details about the equipment can be found in Table 4, with all costs stated in the local currency (COP). Additionally, there is a brief description of the supplementary budget allocated for unforeseen expenses in Table 5.

Table 2: Project's general budget.

	Sources		**TOTAL**
	Author	**UIS**	
Staff	20 000 000	5 000 000	25 000 000
Equipment	-	1 500 000	1 500 000
Materials	70 000	-	70 000
TOTAL	20 070 000	6 500 000	26 570 000

Table 3: Description of staff's budget.

Name	**Title**	**Role**	**Weekly hours**	**Hour Value**	**N. of weeks**	**Resources**		**TOTAL**
						Author	**UIS**	
Javier Ardila	PhD.	Advisor	1	300 000	24	-	7 200 000	7 200 000
Alex Mantilla	MSc(s).	Co Advisor	1	100 000	24	-	2 400 000	2 400 000
Michel Saenz	Undergrad. Student.	Author	20	15 000	24	7 200 000	-	7 200 000
Duvan Gomez	Undergrad. Student.	Author	20	15 000	24	7 200 000	-	7 200 000
TOTAL						14 400 000	9 600 000	24 000 000

Table 4: Description of the used equipment's budget.

Description of the equipment	**Number of weeks**	**TOTAL**
Databases	24	1 500 000
TOTAL		1 500 000

Table 5: Description of additional material's budget.

Description	**Resources**		**TOTAL**
	Author	**UIS**	
Office materials	70 000	-	70 000
TOTAL			50 000

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